

S2P2 vs. SS2P2

Same expressive backbone, two output heads: how the gated bound turns a state-space point process into a *stable* simulator

1 One-line summary

S2P2 is a state-space point process: a stacked latent-linear-Hawkes (LLH) backbone with a *Proj & Softplus* rate head and a rate-coupled mark head. **SS2P2** keeps that backbone *verbatim* and swaps only the two heads for a **gated-bounded (G1) rate** and a **rate-neutral mark head**:

$$\mathbf{SS2P2} = \underbrace{\text{S2P2 LLH backbone}}_{\text{unchanged, expressive}} \rightarrow \underbrace{\text{G1 bounded rate } \lambda(t)}_{\text{Poisson sandwich}} \times \underbrace{\text{rate-neutral } p^*(x | t)}_{\text{decoupled marks}}.$$

The backbone is identical, so any difference in behaviour is attributable to the heads.

2 Shared backbone (both models)

Stacked LLH/SSM layers $\ell = 1, \dots, L$ with diagonal state, ZOH-discretized and parallel-scan computable, with a paper-faithful LayerNorm'd readout producing the embedding $u(t) := u^{(L)}(t) \in \mathbb{R}^H$:

$$x_{t'}^{(\ell)} = \bar{A}^{(\ell)} x_t^{(\ell)} + (\bar{A}^{(\ell)} - I) \tilde{B}^{(\ell)} u^{(\ell-1)} [+ \tilde{E}^{(\ell)} \alpha \text{ at an event}], \quad u^{(\ell)} = \text{LayerNorm}(\text{GELU}(y^{(\ell)}) + u^{(\ell-1)}).$$

Both models read the *same* $u(t)$. They differ only in what comes next.

3 The two heads, side by side

	S2P2 (baseline)	SS2P2 (stable variant)
Rate $\lambda(t)$	$\lambda = s \cdot \text{softplus}(w^\top u + b)$ (<i>Proj & Softplus</i> ; unbounded above)	$h = \sigma(W_o u) \odot \tanh(u) \in (-1, 1)^H$, $\lambda = s \cdot \text{softplus}(w_{\text{eff}}^\top h + b)$, $\ w_{\text{eff}}\ _1 \leq c$ (gated, two-sided bounded)
Mark $p^*(x t)$	$p(k) = \lambda_k / \sum_j \lambda_j$ — <i>coupled</i> to the rate field (or a shared-head softmax)	$p^*(k) = \text{softmax}(\text{MLP}(u))_k$ — <i>decoupled</i> , rate-neutral
$\lambda_x^*(t)$	emergent from the field	$\lambda(t) p^*(x t)$, with $\sum_x p^* = 1$
Rate range	$(0, \infty)$ — can run away in closed loop	$(s \text{ softplus}(b - c), s \text{ softplus}(b + c))$ — finite both sides
Mean calibration	emergent (MLE)	emergent (MLE); s, b set the level — <i>no</i> EMA / branching pin

Why the gate matters (G1). Because $|h_i| < 1$ and $\|w_{\text{eff}}\|_1 \leq c$, we have $|w_{\text{eff}}^\top h| < c$, hence the closed-form two-sided bound

$$\lambda(t) \in (\ell_-, \ell_+), \quad \ell_\pm = s \text{ softplus}(b \pm c) \Rightarrow \text{Poisson}(\ell_-) \preceq N(t) \preceq \text{Poisson}(\ell_+).$$

This is the ‘‘Poisson sandwich’’: in a free-running rollout the event rate can neither explode (no runaway) nor die (no silence), *without* any subcriticality projection. S2P2’s $\text{softplus}(w^\top u + b)$ has no upper bound, so a self-exciting closed loop can diverge. The magnitude lives in the learnable scale s and the dynamic range in $\text{softplus}(b \pm c)$, so the bound is *multiplicative* (wide) rather than additive.

Why decoupled marks. S2P2’s $p(k) = \lambda_k / \sum \lambda_j$ ties the mark distribution to the rate field. SS2P2 reads marks from the same u through an independent softmax MLP, so marks stay fully expressive while never perturbing the bounded total rate ($\sum_x \lambda_x^* = \lambda$).

4 Stability properties

property	S2P2	SS2P2
No runaway in closed-loop rollout	not guaranteed	guaranteed ($\lambda \leq \ell_+$)
No death / silent absorbing state	not guaranteed	guaranteed ($\lambda \geq \ell_- > 0$)
Self-limiting gain	no	yes (tanh saturates, $\text{softplus}' < 1$)
Off-distribution fail-safe	no	yes ($o \rightarrow 0 \Rightarrow \lambda \rightarrow s \text{ softplus}(b)$)
Marks perturb the rate?	yes (coupled)	no (rate-neutral)
Expressive (parallel-scan) backbone	yes	yes (identical)

5 Empirical head-to-head

Identical settings (seq 64 / stride 32, categorical mark head, same `genuine_eval` + `stylized_facts` harness). *Prediction*: top-1 mark accuracy / perplexity over genuine events. *Simulation*: relative error of stylized facts vs. real (lower is better); SIM is their mean.

model	ACC $\uparrow$	PPL $\downarrow$	Fano_re	clus_re	kurt_re	SIM $\downarrow$
SS2P2	0.4519	8.51	0.939	0.241	0.658	0.613
S2P2 (parent)	0.4648	8.47	0.388	1.331	0.159	0.626

Raw stylized facts (real $\rightarrow$ model):

- Volatility clustering ($|r|$ -ACF, lags 1–10): real 0.127 $\rightarrow$ **SS2P2** 0.096, S2P2 0.296 (S2P2 *over-clusters*).
- Excess kurtosis: real 8.79 $\rightarrow$ SS2P2 3.01, **S2P2** 7.39.
- Fano @1s: real 54.9 $\rightarrow$ SS2P2 5.77, **S2P2** 26.0.

Read. Prediction is essentially tied (SS2P2 -1.3 pp ACC, equal PPL): the decoupled softmax head keeps S2P2’s predictive power. Aggregate SIM is tied (0.613 vs 0.626), but the *profile* differs: the G1 bound regularizes the rate, so SS2P2 matches volatility clustering far better while undershooting Fano/kurtosis (the bound plus a tight $\|w\|_1$ cap limit burst magnitude — recoverable with a looser cap / more training). The point SS2P2 buys is **stability**: the bounded rate ran a closed-loop rollout with no blow-up, by construction.

Caveat: these numbers are a local validation on Coinbase-BTC (1 day), 10 epochs on CPU, short rollout — smoke-quality, for isolating the head change. The paper-grade comparison (Gemini data, full config grid, GPU) is staged for the cluster.